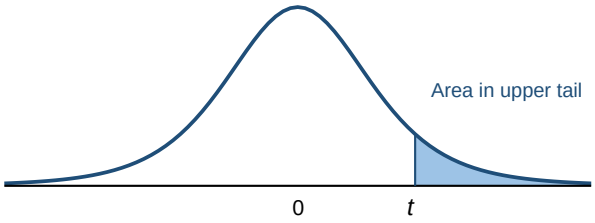
t Distribution Table

Critical values of t for selected upper-tail areas and degrees of freedom

How to read the table. Each entry is the t value that leaves the stated area in the **upper tail**. Find the row for your degrees of freedom (df = n – 1 for a one-sample test) and the column for α.

- **Upper-tail test** (H_a: $\mu > \mu_0$): reject H_0 if $t \geq$ table value.
- **Lower-tail test** (H_a: $\mu < \mu_0$): the curve is symmetric, so use the **negative** of the table value; reject if $t \leq -$(table value).
- **Two-tailed test** (H_a: $\mu \neq \mu_0$): use the column for α/2 (bottom row shows the two-tail α); reject if $|t| \geq$ table value.
- **Example:** df = 99, lower-tail test at α = .05 → critical value = –1.660.
- If your df is not listed, use the next **smaller** df (more conservative).

The ∞ row equals the standard normal z values.



	Area in Upper Tail (one-tailed α)					
df	**.20**	**.10**	**.05**	**.025**	**.01**	**.005**
1	1.376	3.078	6.314	12.706	31.821	63.657
2	1.061	1.886	2.920	4.303	6.965	9.925
3	0.978	1.638	2.353	3.182	4.541	5.841
4	0.941	1.533	2.132	2.776	3.747	4.604
5	0.920	1.476	2.015	2.571	3.365	4.032
6	0.906	1.440	1.943	2.447	3.143	3.707
7	0.896	1.415	1.895	2.365	2.998	3.499
8	0.889	1.397	1.860	2.306	2.896	3.355
9	0.883	1.383	1.833	2.262	2.821	3.250
10	0.879	1.372	1.812	2.228	2.764	3.169
11	0.876	1.363	1.796	2.201	2.718	3.106
12	0.873	1.356	1.782	2.179	2.681	3.055
13	0.870	1.350	1.771	2.160	2.650	3.012
14	0.868	1.345	1.761	2.145	2.624	2.977
15	0.866	1.341	1.753	2.131	2.602	2.947
16	0.865	1.337	1.746	2.120	2.583	2.921
17	0.863	1.333	1.740	2.110	2.567	2.898
18	0.862	1.330	1.734	2.101	2.552	2.878
19	0.861	1.328	1.729	2.093	2.539	2.861
20	0.860	1.325	1.725	2.086	2.528	2.845
21	0.859	1.323	1.721	2.080	2.518	2.831
22	0.858	1.321	1.717	2.074	2.508	2.819
23	0.858	1.319	1.714	2.069	2.500	2.807
24	0.857	1.318	1.711	2.064	2.492	2.797
25	0.856	1.316	1.708	2.060	2.485	2.787
26	0.856	1.315	1.706	2.056	2.479	2.779
27	0.855	1.314	1.703	2.052	2.473	2.771
28	0.855	1.313	1.701	2.048	2.467	2.763
29	0.854	1.311	1.699	2.045	2.462	2.756
30	0.854	1.310	1.697	2.042	2.457	2.750
31	0.853	1.309	1.696	2.040	2.453	2.744
32	0.853	1.309	1.694	2.037	2.449	2.738
33	0.853	1.308	1.692	2.035	2.445	2.733
34	0.852	1.307	1.691	2.032	2.441	2.728
35	0.852	1.306	1.690	2.030	2.438	2.724
36	0.852	1.306	1.688	2.028	2.434	2.719
37	0.851	1.305	1.687	2.026	2.431	2.715
38	0.851	1.304	1.686	2.024	2.429	2.712
39	0.851	1.304	1.685	2.023	2.426	2.708
40	0.851	1.303	1.684	2.021	2.423	2.704
41	0.850	1.303	1.683	2.020	2.421	2.701
42	0.850	1.302	1.682	2.018	2.418	2.698
43	0.850	1.302	1.681	2.017	2.416	2.695
44	0.850	1.301	1.680	2.015	2.414	2.692
45	0.850	1.301	1.679	2.014	2.412	2.690
Two-tail α	*.40*	*.20*	*.10*	*.05*	*.02*	*.01*

Excel equivalents: critical value = `T.INV(1–α, df)` (upper tail) or `T.INV(α, df)` (lower tail); two-tailed = `T.INV.2T(α, df)`; p-value (lower tail) = `T.DIST(t, df, TRUE)`; upper tail = `T.DIST.RT(t, df)`.

t Distribution Table (continued)

	Area in Upper Tail (one-tailed α)					
df	**.20**	**.10**	**.05**	**.025**	**.01**	**.005**
46	0.850	1.300	1.679	2.013	2.410	2.687
47	0.849	1.300	1.678	2.012	2.408	2.685
48	0.849	1.299	1.677	2.011	2.407	2.682
49	0.849	1.299	1.677	2.010	2.405	2.680
50	0.849	1.299	1.676	2.009	2.403	2.678
51	0.849	1.298	1.675	2.008	2.402	2.676
52	0.849	1.298	1.675	2.007	2.400	2.674
53	0.848	1.298	1.674	2.006	2.399	2.672
54	0.848	1.297	1.674	2.005	2.397	2.670
55	0.848	1.297	1.673	2.004	2.396	2.668
56	0.848	1.297	1.673	2.003	2.395	2.667
57	0.848	1.297	1.672	2.002	2.394	2.665
58	0.848	1.296	1.672	2.002	2.392	2.663
59	0.848	1.296	1.671	2.001	2.391	2.662
60	0.848	1.296	1.671	2.000	2.390	2.660
61	0.848	1.296	1.670	2.000	2.389	2.659
62	0.847	1.295	1.670	1.999	2.388	2.657
63	0.847	1.295	1.669	1.998	2.387	2.656
64	0.847	1.295	1.669	1.998	2.386	2.655
65	0.847	1.295	1.669	1.997	2.385	2.654
66	0.847	1.295	1.668	1.997	2.384	2.652
67	0.847	1.294	1.668	1.996	2.383	2.651
68	0.847	1.294	1.668	1.995	2.382	2.650
69	0.847	1.294	1.667	1.995	2.382	2.649
70	0.847	1.294	1.667	1.994	2.381	2.648
71	0.847	1.294	1.667	1.994	2.380	2.647
72	0.847	1.293	1.666	1.993	2.379	2.646
73	0.847	1.293	1.666	1.993	2.379	2.645
74	0.847	1.293	1.666	1.993	2.378	2.644
75	0.846	1.293	1.665	1.992	2.377	2.643
76	0.846	1.293	1.665	1.992	2.376	2.642
77	0.846	1.293	1.665	1.991	2.376	2.641
78	0.846	1.292	1.665	1.991	2.375	2.640
79	0.846	1.292	1.664	1.990	2.374	2.640
80	0.846	1.292	1.664	1.990	2.374	2.639
81	0.846	1.292	1.664	1.990	2.373	2.638
82	0.846	1.292	1.664	1.989	2.373	2.637
83	0.846	1.292	1.663	1.989	2.372	2.636
84	0.846	1.292	1.663	1.989	2.372	2.636
85	0.846	1.292	1.663	1.988	2.371	2.635
86	0.846	1.291	1.663	1.988	2.370	2.634
87	0.846	1.291	1.663	1.988	2.370	2.634
88	0.846	1.291	1.662	1.987	2.369	2.633
89	0.846	1.291	1.662	1.987	2.369	2.632
90	0.846	1.291	1.662	1.987	2.368	2.632
91	0.846	1.291	1.662	1.986	2.368	2.631
92	0.846	1.291	1.662	1.986	2.368	2.630
93	0.846	1.291	1.661	1.986	2.367	2.630
94	0.845	1.291	1.661	1.986	2.367	2.629
95	0.845	1.291	1.661	1.985	2.366	2.629
96	0.845	1.290	1.661	1.985	2.366	2.628
97	0.845	1.290	1.661	1.985	2.365	2.627
98	0.845	1.290	1.661	1.984	2.365	2.627
99	0.845	1.290	1.660	1.984	2.365	2.626
100	0.845	1.290	1.660	1.984	2.364	2.626
∞	0.842	1.282	1.645	1.960	2.326	2.576
Two-tail α	*.40*	*.20*	*.10*	*.05*	*.02*	*.01*